SESSION CHECK

Session 8 — Indian sky, and what to do next

Practical Astronomy Intensive

Name ______________________________ Date ____________________

7 questions **15** minutes

Positional astronomy, not folklore: 360/27 = 13°20′, 12° of elongation per tithi, five limbs from two longitudes. Plus the recognition that Saptarishi is the Dipper they already found — and that the camp has an afterwards.

*Circle one letter for each multiple-choice question. Write short answers on the lines. Tasks marked **Practical** are done, not written — your assessor will watch.*

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1 On night one you found the Big Dipper and Polaris. Tonight someone mentions Saptarishi and Dhruva. What are they pointing at?

- a A different pole star from Polaris, used in the Indian system
- b The same two things — Saptarishi IS those seven stars, and Dhruva IS Polaris; only the names and the stories differ
- c Two different patterns in another part of the sky, which you have not found yet
- d The Sanskrit translations of the Greek names, coined after Greek astronomy reached India

2 The ecliptic gets divided into 12 rashis and also into 27 nakshatras. What is a nakshatra, precisely?

- a One of 27 divisions chosen because 27 is a traditionally auspicious number
- b A fixed 13°20′ segment of the ecliptic — 360/27 — cut to the Moon's daily step of about 13° against the stars, and named for a marker star (yogatara) that lies in it
- c A constellation, whose boundaries follow the shape of the star pattern it is named for
- d The 12 rashis renamed in Sanskrit — the same divisions under different labels

3 A tithi is often glossed in English as 'lunar day'. How long is one, and what sets its length?

- a Exactly 24 hours — tithi is simply the Indian word for a day
- b Exactly one thirtieth of 30 days, since 30 tithis make a month
- c Exactly 12 hours, one for each degree of elongation gained
- d Anywhere from about 19 to 26 hours — a tithi ends when the Moon has gained 12° of elongation on the Sun, and the Moon's speed varies around its elliptical orbit

4. A panchanga lists five quantities each day: vara, tithi, nakshatra, yoga, karana. What is the document?

   a. A computed almanac — all five limbs are functions of the Sun's and the Moon's longitudes on that day, which is to say an ephemeris in cultural form
   b. An astrological text, since that is what people consult it for
   c. A table copied forward from classical texts, preserved unchanged for accuracy
   d. Five separate traditions bundled into one book over the centuries

5. The camp ends tomorrow. A student owns binoculars, no telescope, and lives under a suburban sky. What can she realistically contribute to actual research?

   a. A great deal — AAVSO variable-star estimates from binoculars, IMO meteor counts by naked eye, occultation timings, or Zooniverse classifications all feed straight into professional work
   b. She can observe for her own enjoyment, but amateur numbers are a hobbyist exercise nobody really uses
   c. Very little on her own — one observer's handful of estimates cannot matter against professional data
   d. Nothing yet — she needs a telescope and a CCD before her measurements would be worth submitting

6. Arundhati–Vasishtha is the classic naked-eye test in the Indian sky. Which pair of stars is it, and where do you look?

   a. Two stars of the Dipper's bowl, named as a pair only after telescopes revealed them
   b. Dhruva and its faint companion, near the celestial pole
   c. A pair in Orion, corresponding to the Greek naming of the belt
   d. Mizar and Alcor — the middle star of the Big Dipper's handle, which resolves into two for anyone with good eyes and a dark sky

7. A panchanga shows a tithi that begins after sunrise and ends before the next sunrise, so no sunrise falls inside it — that date's tithi is skipped (kshaya). How is that possible?

   a. The Moon was invisible that day, so the tithi could not be observed and was dropped
   b. Tithis are all 24 hours, so the skip must come from a leap-day adjustment in the civil calendar
   c. It is a computational error in that panchanga — a date always carries exactly one tithi
   d. Tithis vary in length, so a short one (near 19 hours) can start and finish entirely between two sunrises, while a long one (near 26 hours) can span two sunrises and repeat (adhika)